Automatic Image Colorization using SVR and MRF (ICM) Smoothing

UE24CS352A – Machine Learning · Mini-Project · Problem 150 · Based on Ho & Ramesh, “Automatic Image Colorization” (CS229)

1. Problem Statement

Colorization is an **ill-posed** problem: many different colours can map to the same gray value, so a grayscale pixel alone does not determine its colour. The goal of this project is **fully automatic colorization of nature/landscape photographs** – with no user scribbles and no reference image. The model receives only the luminance channel **Y** and must predict the chrominance channels **U** and **V**; these are recombined with Y and converted back to RGB.

2. Dataset Details

Property	Value
Content	Colour landscape photographs (sky, mountains, water, fields) – a Yellowstone-style nature setting as in the paper
Size	215 images, rescaled to a constant width of 500 px
Split	80 / 20 random split: **172 train / 43 test** images (seed 42)
Validation	15% of training images (26) held out only for hyperparameter tuning; final SVRs are trained on all 172
Cleaning	Black letterbox bars auto-cropped; images that are already grayscale are removed
Training samples	≈39,330 superpixels (≈250 per image); targets are the mean U and mean V of each superpixel

3. Approach

Rather than predicting colour for every pixel, the image is divided into superpixels and colour is predicted per segment, which lets colours follow image structure and keeps the problem tractable. The pipeline has six stages:

1. YUV conversion	2. SLIC superpixels	3. FFT features	4. Two SVRs	5. MRF + ICM	6. Recombine
Model sees only Y	~250 segments computed on Y	10×10 patch at each centroid	RBF kernel: Y-features → U and V	Smooth neighbouring segments	Y + U,V → RGB

Local estimate (SVR). The magnitude of the 2-D FFT of the 10×10 patch around each superpixel centroid gives 100 texture features, supplemented by 4 context features. Two RBF-kernel Support Vector Regressors (one each for U and V) map these features to the segment’s mean chrominance.

Global smoothing (MRF + ICM). Each channel is modelled as a Markov Random Field over adjacent superpixels. The energy of segment *i* combines a Gaussian term pulling it towards its SVR prediction and a smoothness term towards its neighbours (only neighbours with similar texture features are linked). Iterated Conditional Modes minimises it with a closed-form update, repeated until convergence.

4. Implementation Overview

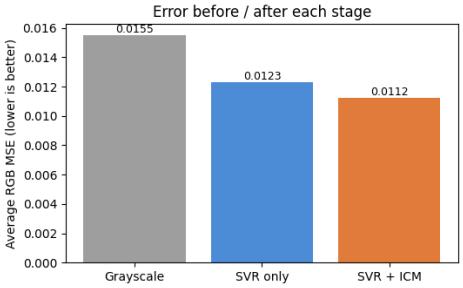
Implemented in Python in a single Jupyter notebook using **NumPy, SciPy, scikit-image** (SLIC, YUV conversion), **scikit-learn** (SVR, StandardScaler) and joblib (model persistence). Key details:

- **Features:** log-compressed FFT magnitudes (100) + vertical position, segment mean/std Y and image mean Y (4). All features are standardised; the context features are weighted ×3 so they are not swamped by the FFT bins.
- **SVR tuning:** grid search over C and ε on the validation images selected **C = 0.125, ε = 0.01** for both U and V (the paper used ε = 0.0625). Final fit takes several minutes on CPU.
- **MRF tuning:** γ, σ and the neighbour-similarity percentile were also tuned on validation data; the chosen values (γ ≈ 128, σ ≈ 0.1, percentile 90) smooth far more strongly than the paper’s γ = 2.
- **Saturation boost:** predicted U and V are multiplied by 1.4 before conversion to RGB to counter washed-out colours.
- **Evaluation:** per-image RGB mean squared error (MSE) and mean per-pixel Euclidean RGB distance on the 43 test images, compared across three stages: grayscale, SVR only, and SVR + ICM.

5. Results

Error falls at every stage of the pipeline (RGB values scaled to [0, 1]; lower is better):

Stage	Avg. RGB MSE	Mean RGB distance
Grayscale input (baseline)	0.0155	0.164
SVR only	0.0128 (–17.6%)	–
SVR + ICM (final)	**0.0112** (a further –10.3 pts)	**0.138**



- SVR alone lowers MSE by **17.6%** relative to grayscale; ICM smoothing removes a further **10.3 percentage points**.
- ICM improved **84% of test images** over the SVR-only output.

Qualitative results on test images



Figure: columns (left → right) – original, grayscale input, SVR only, SVR + ICM. Sky, water, grass and mountains receive plausible, distinct colours; ICM removes the isolated colour blotches visible in the SVR-only column.

6. Conclusions

- Automatic colorization learned from a training corpus is feasible with **classical machine learning**: SLIC superpixels, FFT texture features and SVR regression produce plausible colours without any user input.
- **MRF/ICM smoothing consistently helps**: test MSE falls from 0.0155 (gray) to 0.0112, and 84% of test images improve over SVR alone. Spatial coherence matters as much as the local regressor.
- **Limitations:** output is muted and biased towards blue (warm yellows and browns are under-represented); colour bleeds across segment boundaries; one colour per segment averages plausible alternatives and lowers saturation; the dataset is small (215 landscape images).
- **Future work:** predict colour *distributions* (multimodal) instead of a single value; use a pixel-wise MRF to reduce bleeding; add CNN features for global context; adopt CIELAB colour space; train on larger, more varied data.